

"From bathroom constraints to a personalized, space-aware bathroom design."

The Problem

- Bathroom product selection is constrained all at once by physical space, budget, style, and product compatibility.
- Customers and sales assistants juggle these manually, with no tool that validates a bundle against all four together.
- Generic AI chat tools can suggest products fluently -- but can't guarantee the suggestion actually fits the room or the budget.

The Solution

- An AI-assisted design workflow that converts natural-language customer requirements into a validated bathroom product bundle.
- Generative AI interprets language and explains results; a deterministic engine validates space, budget, and feasibility.
- Output: a scored product bundle, a labeled 3D-style isometric visualization, a water-use estimate, and a plain-language explanation -- grounded, not guessed.

Generative AI proposes. Deterministic optimization validates.



■ AI-driven ■ Deterministic (source of truth)

AI is responsible for:

- Understanding natural-language requirements
- Extracting structured preferences (Pydantic-validated)
- Generating grounded, plain-language explanations

Deterministic engine is responsible for:

- Prices, dimensions, and physical fit
- Budget, category, and clearance constraints
- Bundle optimization and sustainability calculations

Tech stack

Python · Streamlit · Pydantic · OpenAI API (structured outputs) · Matplotlib · python-dotenv · pytest · JSON product catalog

Demo scenario

Bathroom: 8 × 6 ft

Budget: Rs. 2,50,000

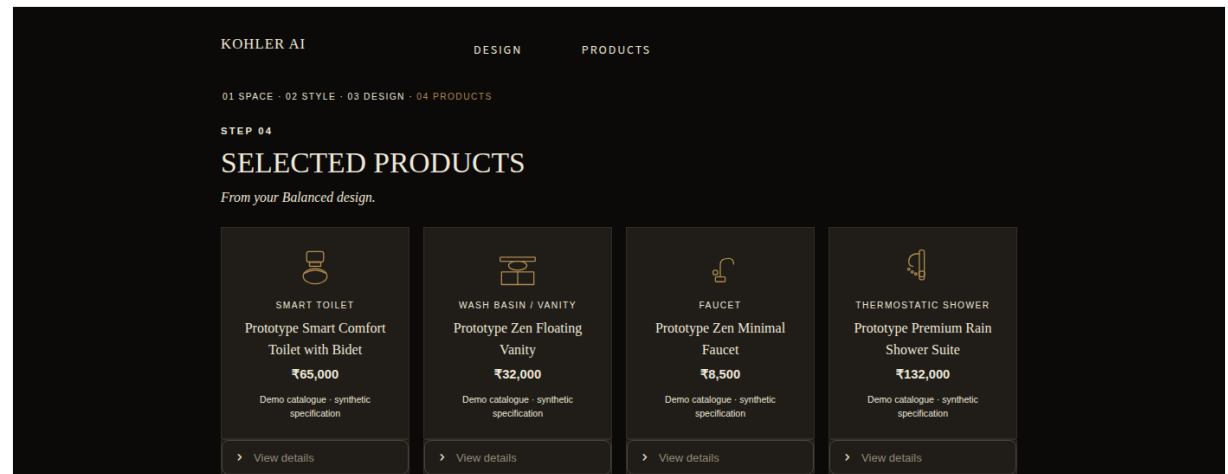
Style: Japanese Zen

Required: smart toilet, faucet,
thermostatic shower, vanity

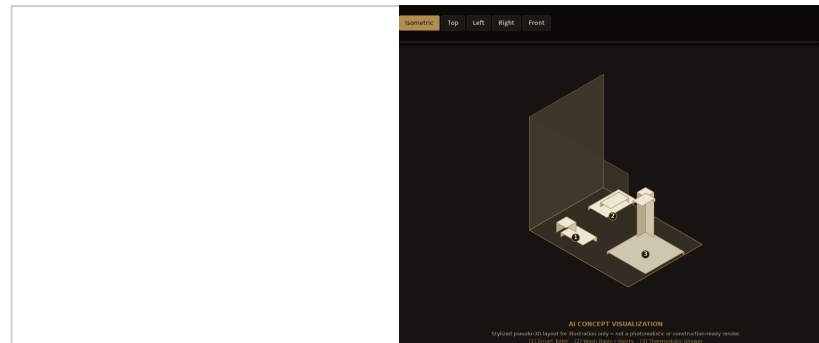
Innovation

**“Generative AI proposes.
Deterministic optimization validates.”**

The system never lets the LLM invent feasibility, dimensions, or pricing -- every claim traces back to deterministic Python code.



Actual Streamlit output — selected products (offline mode, no API key)



Labeled, multi-angle 3D-style isometric visualization (Isometric/Top/Left/Right/Front)

Business Value

- Personalized customer experience at scale
- Faster product discovery for sales teams
- Space-aware recommendations, not guesswork
- Transparent trade-offs (Essential/Balanced/Premium)
- Never a dead end -- a labeled best-effort design is offered even when no bundle fits the space or budget exactly
- Scalable digital design assistance

Sustainability

- Estimated annual water-use analysis per bundle
- Usage-based comparison vs. a baseline fixture
- Water-efficiency is a scored optimization input
- Every figure labeled as an estimate, never a claim
- Aligned with KOHLER's water-conservation focus

Future Scope

- Official KOHLER catalogue integration
- Real-time product availability & pricing
- Image / floor-plan understanding
- Photorealistic 3D rendering & AR walkthrough
- Plumbing-aware layout validation
- Installation/service integration

“AI doesn't replace bathroom design expertise — it makes personalized design scalable.”